

## Appendix B: Technical Notes

## Determinism I

All stochastic steps use fixed seeds (seed = 42 for NMF, SVD, and graph layouts). The fixture corpus, TF-IDF/SVD embeddings, and topic factorization are byte-stable across runs. Graph centrality scores are rounded to a fixed precision and ranked with a node-id tiebreaker so that floating-point non-associativity in iterative solvers cannot perturb the reported rankings. Record identity uses a content digest (SHA-1, `usedforsecurity=False`) rather than a salted hash, so de-duplication and corpus byte-stability hold across processes.

Each record is a Paper dataclass with: title, abstract, authors (list of Author), year, DOI, arXiv ID, Semantic Scholar ID, OpenAlex ID, venue, citation count, references (list of canonical IDs), publication date, PDF URL, open-access flag, and full-text source. The canonical identifier hierarchy governs de-duplication and citation resolution:

$$\text{canonical\_id} = \begin{cases} \text{doi: + normalize(DOI)} & \text{if DOI present} \\ \text{arxiv: + arXiv\_id} & \text{if arXiv ID present} \\ \text{s2: + S2\_id} & \text{if S2 ID present} \\ \text{openalex: + OpenAlex\_id} & \text{if OpenAlex ID present} \\ \text{title: + SHA1(title)[: 16]} & \text{otherwise} \end{cases}$$

DOI normalization lower-cases the DOI and strips any `https://doi.org/` or `dx.doi.org/` prefix, so the same paper returned by two engines under different case or format variants merges to a single canonical ID.

Non-negative matrix factorization decomposes the TF-IDF document-term matrix  $\mathbf{V} \in \mathbb{R}^{m \times n}$  (where  $m$  is the number of documents and  $n$  is the vocabulary size) into  $\mathbf{W} \in \mathbb{R}^{m \times k}$  and  $\mathbf{H} \in \mathbb{R}^{k \times n}$ , where  $k$  is the number of topics (here 5). The factorization minimizes:

$$\min_{\mathbf{W}, \mathbf{H} \geq 0} \|\mathbf{V} - \mathbf{WH}\|_F^2$$

using multiplicative update rules (Lee and Seung 1999) with a fixed random seed for reproducibility. The topic-term matrix  $\mathbf{H}$  gives the top terms per topic; the document-topic matrix  $\mathbf{W}$  gives each document's topic loadings.

## Growth Rate Estimation I

The compound annual growth rate is:

$$\text{CAGR} = \left( \frac{N_{\text{end}}}{N_{\text{start}}} \right)^{1/(T_{\text{end}} - T_{\text{start}})} - 1$$

where  $N_{\text{start}}$  and  $N_{\text{end}}$  are the publication counts in the first and last years of the corpus, respectively. The doubling time is  $t_d = \ln(2)/\ln(1 + \text{CAGR})$ . For this run: CAGR = 5.48%, doubling time = 9.2 years.

## Configuration Surface I

A single `manuscript/config.yaml` controls the search term, per-engine query and keyword sets, engine enable toggles, subfield taxonomy, hypotheses, full-text and embedding options, and paper metadata. This run drew on 10 engines, a 6-bucket taxonomy, and 6 hypotheses.

## Artifacts I

Intermediate and final outputs live under output/ and are disposable and regenerable:

# Artifacts I

| File                             | Stage | Description                                                                                                  |
|----------------------------------|-------|--------------------------------------------------------------------------------------------------------------|
| corpus.jsonl                     | 01    | De-duplicated corpus (2334 records)                                                                          |
| temporal_analysis.json           | 02    | Year counts, CAGR, doubling time, peak year                                                                  |
| subfield_classification.json     | 02    | Per-bucket paper counts                                                                                      |
| subfield_timeline.json           | 02    | Per-subfield annual breakdown                                                                                |
| tfidf_data.json                  | 02    | TF-IDF matrix, feature names, doc tokens                                                                     |
| topics.json                      | 02    | NMF topic-term distributions                                                                                 |
| citation_network.json            | 02    | Network metrics, PageRank, HITS, communities                                                                 |
| citation_graph.gml               | 02    | GraphML citation graph                                                                                       |
| nanopublications.jsonl           | 03    | LLM-extracted assertions (0 in this run)                                                                     |
| hypothesis_scores.json           | 03    | Per-hypothesis evidence scores                                                                               |
| fulltext_assessment.json         | 06    | Abstract/OA/PDF coverage report                                                                              |
| fulltext/fulltext_inventory.json | 06    | Provider metadata, declared licenses, and local PDF SHA-256 checksums; no license is inferred from OA status |